SIMATS ENGINEERING

# SIMATS Online Programming Lab

JAVA PROGRAMMING

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## CO2-AT2-Assignment

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QUESTIONS

Q1
Bank Token Queue Management System
10 marks

Q2
Student Email Username Extractor
10 marks

Q3
Warehouse Damaged Item Removal
10 marks

3/3 answered

### Q1 Bank Token Queue Management System

10 marks

A bank serves customers based on token order. Using Queue:

Add customer tokens using offer()
Serve the next customer using poll()
Display the upcoming customer using peek()

**Your Code**

```java
import java.util.LinkedList;
import java.util.Queue;

public class BankQueue{
    public static void main(String[] args){
        Queue<Integer> queue=new LinkedList<>();
        queue.offer(1);
        queue.offer(2);
        queue.offer(3);
        queue.offer(4);
        System.out.println("served Customer: " + queue.poll());
        System.out.println("upcoming Customer: " + queue.peek());
        System.out.println("remaining Queue: " + queue);
    }
}
```

**Input**

stdin input

**Output**

```
served Customer: 1
upcoming Customer: 2
remaining Queue: [2, 3, 4]
```

Run Save

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- **Q1** Bank Token Queue Management System — 10 marks
- **Q2** Student Email Username Extractor — 10 marks
- **Q3** Warehouse Damaged Item Removal — 10 marks

3/3 answered

### Q2 Student Email Username Extractor

10 marks

A college portal stores student emails such as arun23@college.edu. Develop a program that:

Finds the position of @ using indexOf()

Extracts the username using substring()

Displays the username length using length()

**Your Code**

```java
public class StuEmail{
    public static void main(String[] args){
        String email="arun23@college.edu";
        int position=email.indexOf("@");
        String username=email.substring(0, position);
        System.out.println("Position @: " + position);
        System.out.println("Username: " + username);
        System.out.println("Length: " + username.length());
    }
}
```

**Input**

stdin input

**Output**

```
Position @: 6
Username: arun23
Length: 6
```

Run Save

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QUESTIONS

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Q3
Warehouse Damaged Item Removal
10 marks

3/3 answered

## Q3 Warehouse Damaged Item Removal

10 marks

A warehouse stores item names in an inventory list. Damaged items must be removed safely while traversing the collection.

Requirements:

Store inventory items in an ArrayList
Traverse the list using an Iterator
Remove items marked as Damaged and display the remaining inventory

**Your Code**

```
import java.util.ArrayList;
import java.util.Iterator;

public class Inventory{
    public static void main(String[] args){
        ArrayList<String>items=new ArrayList<>();
        items.add("Laptop");
        items.add("Damaged");
        items.add("Keyboard");
        items.add("Damaged");
        items.add("Mouse");
        items.add("Monitor");
        Iterator<String> it = items.iterator();

        while(it.hasNext()){
            String item=it.next();
            if(item.equals("Damaged")){
                it.remove();
            }
        }
        System.out.println("Remaining Inventory:");
        for(String item:items){
            System.out.println(item);
        }
    }
```

**Input**

stdin input

**Output**

```
Remaining Inventory:
Laptop
Keyboard
Mouse
Monitor
```